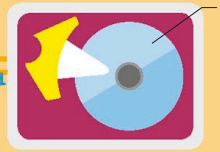
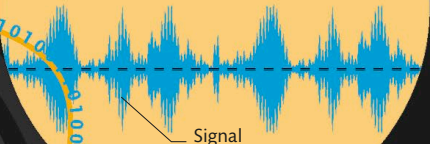
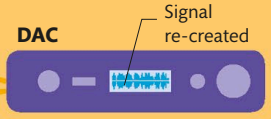
**5 Storing sound**
The sequence of binary numbers can be stored in the device's memory—on a hard disk or a USB flash drive, for example.



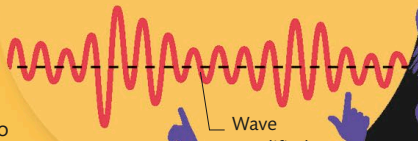

**6 Re-creating sound**
For playback, a processor retrieves the sequences from storage, ready to re-create the audio signal.




**7 Back to analogue**
A digital-to-analogue converter (DAC) uses the sequence of binary numbers retrieved from storage to re-create the audio signal.

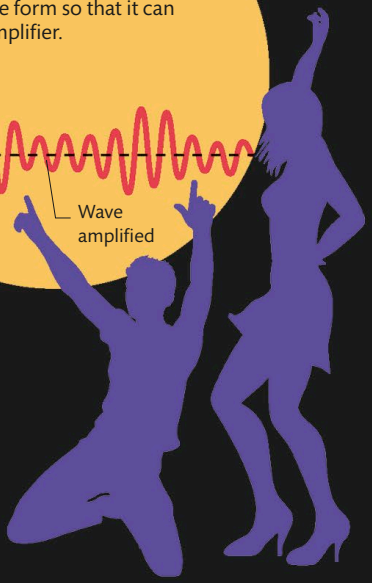



**8 Amplifying the signal**
The signal is now restored to analogue form so that it can drive an amplifier.



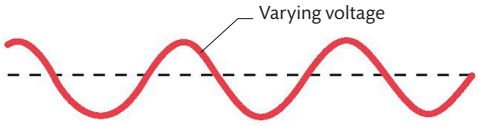

**9 Playback**
The amplified audio signal pushes a cone inside the speaker back and forth, creating sound waves of varying pressure.



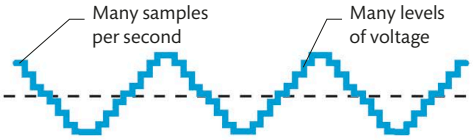


## Sound quality

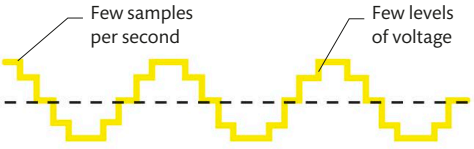
The quality of digital sound depends on how many samples are used per second and how many bits (binary digits) are used to represent the number for each sample. The quality of sound on compact discs is standardized. It uses 44,100 samples per second and 16 bits per sample.




**Original analogue audio signal**
The audio signal the microphone produces is a smooth wave of varying voltage. It goes up and down hundreds or thousands of times each second.




**Good quality**
Digital sound cannot re-create a perfect audio signal, but the more levels of voltage, and the more samples per second, the better.




**Poor quality**
Poor-quality audio is choppy and distorted, because it has fewer bits per sample—which means fewer levels of voltage—and fewer samples per second.

## ON THE PHONE

When you speak on the phone, the sound of your voice travels across the telephone network in digital form. A smartphone has an ADC and a DAC built in. For landline phones, the ADC and DAC are outside the home.

